

A cyber-physical dynamic control approach for smooth traffic and clean urban networks using a bi-objective regulation function at signalized intersections

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Abstract—Traffic flow rising in urban areas makes road networks more congested. This leads to delays, slow-downs and length queuing of vehicles. These issues impact directly the environment, the cities atmosphere, and harms society as well. Therefore, in order to minimize these negative impacts, urban traffic requires adequate regulation methods. To this end, we propose in this paper an adaptive traffic light control technique at signalized intersections. It is a dynamic control approach where the number of vehicles and their arrival times at urban intersections are considered for decision-making regarding lights switching. The dynamic behavior of the studied physical system can be seen as a discrete events system. As powerful modeling and analysis tool we use Timed-Synchronized-Petri-Net (TSPN). The technique of "Master-Slaves" is adopted in this approach. In fact, a "master" reactive controller takes care of the evacuation of the vehicles from the intersection and the switching of the lights "slaves". These latter are represented by TSPN modules. The proposed control approach is based on the technique of Cyber-Physical systems where the digital model of the urban intersection (physical entity) is modeled and simulated using SUMO simulator. Thanks to the developed formal and simulation models, some interesting network's properties are analysed and checked. The proposed solution allows for a smoother traffic flow in the studied intersection, reducing energy consumption and environmental impact.

I. INTRODUCTION

The rapid progress in traffic movements across all forms of transport is attributed to industrial prosperity and the population increasing. As a consequence, different traffic management challenges and issues are gradually emerging. Traffic congestion is one of them and deemed the most popular problem for transport operators and research community. Traffic congestion is mainly caused by huge number of vehicles circulating in urban areas. This huge number leads to queuing issues and, subsequently, to vehicles delays. Unfortunately, within the entire network, and in particularly at the intersections, the vehicles get sluggish and create traffic congestion.

Strategies for urban traffic management are a technical field with long-term strategic activities and comprehensive studies. In order to address the above-mentioned drawbacks, transportation operators and scientific researchers developed various control and management schemes.

The most common solution for managing traffic and evacuating automobiles as securely and smoothly as possible is traffic lights at signalized intersections. The authors of [1]

categorized optimization methods and road traffic management into pretimed, traffic response, and predictive policies. The first process is the most one used now as traffic management technique. In order to control traffic at an intersection, the authors of [2] have proposed cyber-physical multi-agent framework using predetermined timings of traffic lights. A second category includes responsive traffic techniques that use scheduling plans for adaptive traffic signals. Indeed, these methods are based on the optimization and synchronization of traffic signals online. This category of complex methods monitor the traffic lights at the intersections as explained in [3]. Indeed, Internet of things (IoT) paradigm and cyber-physical technique are used where sensors track and collect the data of traffic at the intersection (physical entity). These data are then sent to a virtual controller (Master) in real time. After processing these data and making a decision, the controller comeback to the intersection with a control method for managing adequately and smoothly traffic within the intersection. This paper belongs to this control approach based on a bi-objective traffic regulation used by the controller (master).

In [4], intersection traffic signal management is studied as a discrete event system (DES). One of the numerous mathematical and graphical modelling methods for describing traffic control networks as DESs is Petri Nets (PNs). The robustness of PNs for modeling concurrent and dynamic operations in DES [5] has been proven in many research works in the literature. Moreover, in a straightforward way, its graphical representations help model (with graphic notations) certain complicated information in complex structures. First, PNs was added to the smart urban traffic network model in [6]. Time is a primary requirement in road traffic structures; ordinary PNs do not deal with this problem. Thus, [7] model the exchanging of traffic lights, utilizing the timed Petri Nets category. The traffic signals duration in that paper are focused on pretimed periods.

The literature review in [8] shows multiple classes usage of PNs in urban traffic modelling, study and control. Indeed, in order to model traffic light regulation for an emergency situation, deterministic and stochastic PNs are also effectively used in [9].

Recently, Synchronized Timed Petri Nets (STPN) are used in [10] to face road traffic congestion. The research work suggested a legislative traffic light management strategy with a fixed traffic signal timing plan that prohibits vehicles from approaching congestion zones on the network.

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An architecture of traffic flow and control logic are suggested using mathematical programming techniques in [11] and [12]. In addition, some artificial intelligence methodologies are proposed in [13]. Unfortunately, they pose a major difficulty of transparently portraying the transitions of the traffic light phases. In addition, they struggle to deal with the question of how traffic lights management logic may be applied. The authors of [10] have revealed this downside can be handled using a STPN model. Even so, by fixed timing plans, they fail to maximize the efficiency of urban traffic networks.

Considering these related studies (benefits and limits) and in order to bring our contribution in this field, we concentrate our efforts by developing a control approach for traffic signals using modular STPN models. A sensitive traffic algorithm is therefore developed to monitor the synchronization of traffic signals. This will enable to evacuate properly vehicles from the intersection and maximize the network efficiency by decreasing vehicle queuing times, fuel consumption, and overall gas pollution at urban area. In addition, using a DES simulation based on SUMO simulator, we apply the suggested control logic on virtual network of physical intersections by adopting the technique of cyber-physical systems. Then, we perform some simulations and analyse obtained results.

Petri nets (with theoretical models) and SUMO (with simulation and virtual models) are used in complementary way to check and validate the proposed approach. Indeed, Petri nets allow to model the behavior of the system and represent all its components interaction (controller, traffic status, signalized lights). The simulation of the system behavior helps to visualize, observe, check and validate the network functioning. Thanks to this powerful simulator (SUMO), some qualitative parameters and system information will be evaluated, such as vehicles waiting time, gas emissions, fuel consumption, etc. In addition, we use the SUMO simulation to see the effect of our proposed strategy against developed strategies in the literature. In other words, PN tool is used for representing and studying the theoretical system behavior, and SUMO enables to illustrate the real functioning of the system and extract the parameters required to perform the whole analysis. The finding of this analysis will be then integrated into the physical intersection in order to reach targeted objectives in traffic regulation.

The remainder of this paper is organized as follows. Section 2 offers the problem statement and modeling methodology. The developed dynamic strategy for regulation is discussed in section 3. In section 4, the simulations results and discussions are recorded. The last section address a conclusion and some perspectives.

II. PROBLEM STATEMENT AND MODELLING APPROACH

In this analysis, we consider the most generalized intersection. It may be adapted to any real intersection by adding or removing a movement(s) to one or more of its lanes. In this study, eight movements are taken into account for each intersection. These movements are: $\langle Ns \rangle$ (North-straight-ahead),

$\langle Nl \rangle$ (North-left-turn), $\langle Es \rangle$ (Est-straight-ahead), $\langle El \rangle$ (Est-left-turn), $\langle Ss \rangle$ (South-straight-ahead), $\langle Sl \rangle$ (South-left-turn), $\langle Ws \rangle$ (West-straight-ahead), $\langle Wl \rangle$ (West-left-turn). In each movement, installed sensors at a specified distance from the stop lines offer data of traffic flow.

The subclass of PNs used for modeling traffic flows and lights is represented in this subsection. More precisely, we use a Timed Synchronized Petri Net (TSPN) which is a bipartite graph defined by the eight-tuple $TSPN = (P, T, A, f_w, H, M_0, \tau, Evt)$:

- P : is a finite set of discrete places.
- $T = T_s \cup T_t$ is a finite set of transitions with synchronized transitions subset T_s and temporized transitions subset T_t .
- $A \subseteq (P \times T) \cup (T \times P)$ is a finite set of arcs.
- $f_w : A \rightarrow \{1, 2, \dots\}$ is the arc-weight function.
- $H \subseteq T \times P$ a set of inhibitor arcs from P to T.
- $M_0 : P \rightarrow \{0, 1, 2, \dots\}$ is an initial marking.
- $\tau : T \rightarrow R^+$ firing time function, where R^+ is the set of non negative real numbers.
- $Evt = \{C_{\langle ij \rangle}, D_{\langle ij \rangle}\}$ is a set of external events associated with certain transitions. In this notation, the index i represents directions: north, south, east, and west, respectively ($i \in \{N, S, E, W\}$); and j represents straight-ahead and left-turn, respectively ($j \in \{s, l\}$).

The global and generalized TSPN model of each direction and each movement is illustrated in Fig.1. It is composed of eight sub-models, with $\langle ij \rangle = (\langle Ns \rangle, \langle Nl \rangle, \langle Es \rangle, \langle El \rangle, \langle Ss \rangle, \langle Sl \rangle, \langle Ws \rangle, \langle Wl \rangle)$. Each TSPN sub-model contains six places and six transitions. Temporizations and external conditions in this model control the frequency and occurrence dates of the events. Fig.2 illustrates the definition of PN's places and transitions (for more details see [3]).

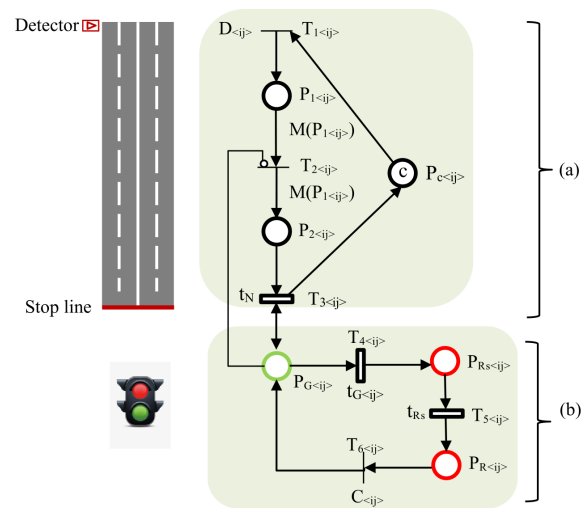


Fig. 1: Generic TSPN model for each movement $\langle ij \rangle$: (a) Sub-model of traffic flow. (b) Sub-model of traffic signals.

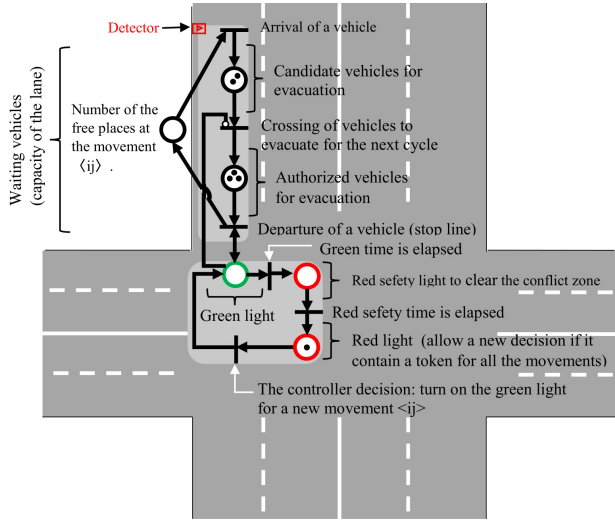


Fig. 2: Definition of PN's places and transitions.

III. BI-OBJECTIVE FUNCTION BASED CONTROL STRATEGY

The key purpose of the master (an adaptive controller) is to find for each cycle the priority movement(s) to be served while switching on to green their signal lights, and avoid conflicts and disagreements. Also, the master updates and adjusts the duration of green light for each selected movement. Through this study, we intend to fulfill these four requirements by regulating road traffic into the studied network: 1) Handle and manage conflicts between various movements to offer safety in the conflict area. 2) Reduce the queuing length at the network. 3) Reduce vehicles' delays and ensure smooth traffic at the network. 4) Decrease environmental impacts (gas emission and fuel consumption). To make a decision, the master bases on these two criteria for each movement at the intersection: the arrival time and the number of waiting vehicles. Based on both criteria, a bi-objective function F (eq.4) is suggested in order to make the adequate and relevant decision. To this end, we define required parameters for implementing this bi-objective function. We define then the vector N that symbolizes the number of vehicles in the movements (eq.2). The vector T represents the arrival time of the vehicle that has stayed the longest in each movement (eq.1). The matrix E contains the possible combinations of movements (eq.3). However, each line of the matrix E represents a combination of two movements and each column represents a movement (m movements). In each line of the matrix E , the two components with the value 1 correspond to the two compatible movements and other components, whose value is 0, corresponds to all incompatible movements (with conflicts).

$$T = (t_1 t_2 \dots t_i \dots t_m)^T, \text{ with } t_l = \min_{1 \leq k \leq N(l)} t_{al}(k) \quad (1)$$

$$N = (n_1 n_2 \dots n_i \dots n_m)^T \quad (2)$$

with m is the number of movements ($m = 8$ in our case), l refers to the movement ($1 \leq l \leq m$), $t_{al}(k)$ is the arrival time of the vehicle k to the movement l , and $N(l)$ is the number of vehicles at the movement l .

$$E = \begin{bmatrix} E_1 \\ E_2 \\ \vdots \\ E_i \\ \vdots \\ E_c \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (3)$$

$$F(N, T, \alpha, \beta) = \alpha \max_{1 \leq i \leq c} \sum_{j=1}^m E(i, j) \cdot N(j) + \beta \min_{1 \leq j \leq m} T(j) \quad (4)$$

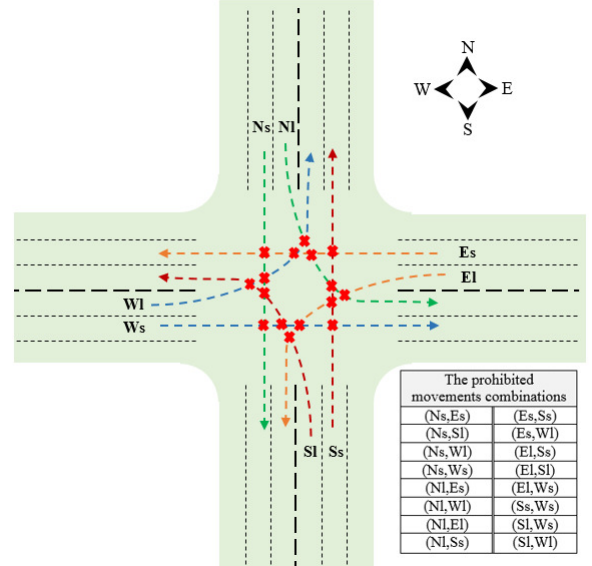


Fig. 3: Prohibited combinations of movements.

with, $N \in R^{m \times 1}$ (eq.2), $T \in R^{m \times 1}$ (eq.1), the variables α and $\beta \in \{0, 1\}$, c is the amount of possible combinations, m is the movements number, and $E \in R^{c \times m}$ (eq3).

The proposed algorithm for the control technique suggested in this work is given in Fig.4. Regulation begins with the red traffic signal over all movements. Firstly, we put all possible combinations of two movements in the matrix E (eq.3). Then, the prohibited combinations (see Fig.3) are removed from E . A prohibited combination represents two movements with conflict at the intersection. For example, if the green light is given simultaneously to the movements (N_s, E_s) a dangerous situation is generated in the conflict zone with a potential occurrence of accidents. So, the matrix E still contains just the safe combinations. As a real-time algorithm, the number of vehicles list for all movements N is updated before making a decision by the master. The variable parameters α and β take zero or one as values according to a test about the number of vehicles state and their arrival

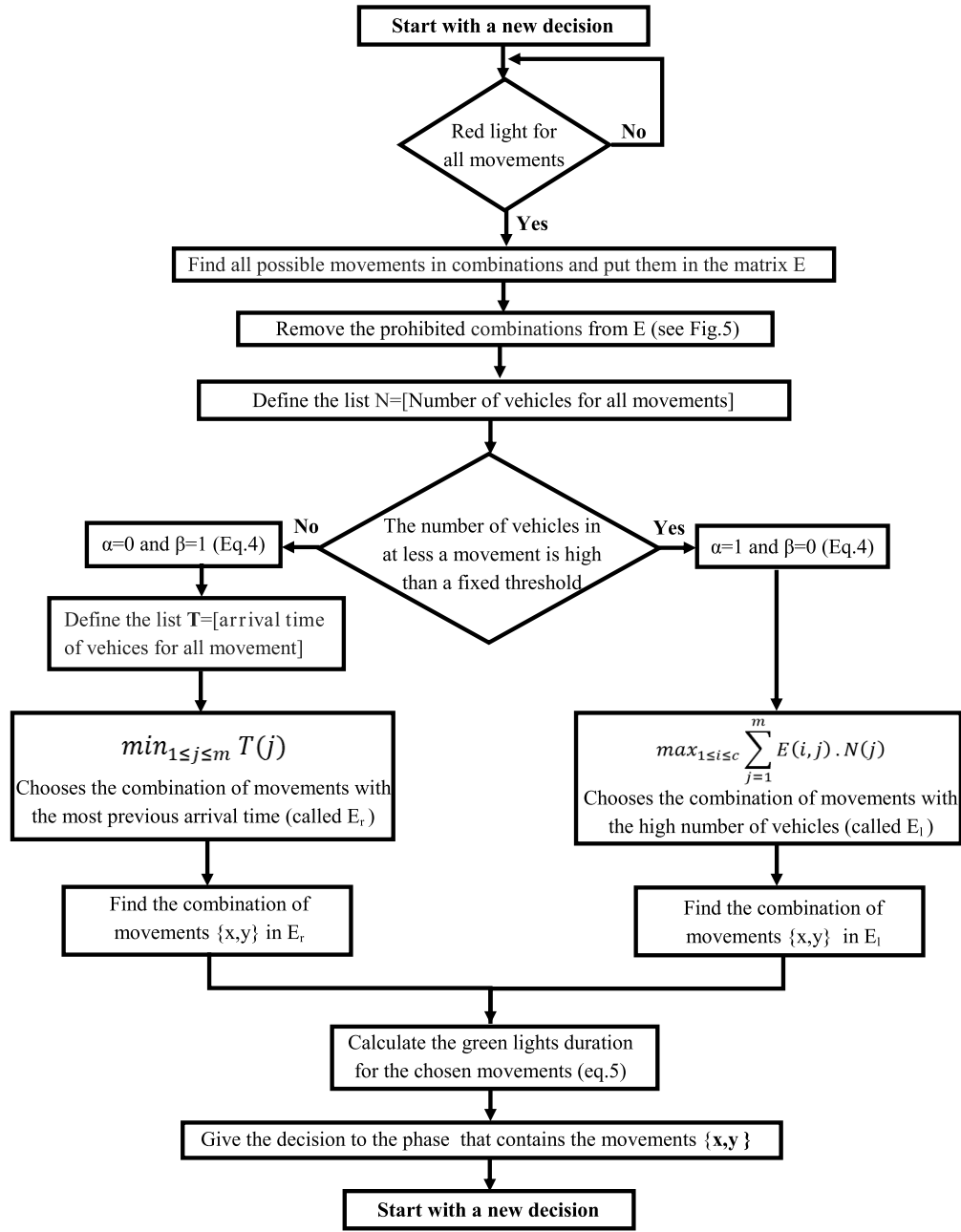


Fig. 4: Bi-objectives based control strategy.

times. We consider two cases: if $\alpha = 1$ and $\beta = 0$, the controller decision is taken according the number of vehicles into the movements (Two movements without conflicts from the matrix E with highest number of vehicles are chosen (the line E_l is given in the algorithm). Otherwise, if $\alpha = 0$ and $\beta = 1$, the master selects from the matrix E the combination of movements E_r with the most previous vehicles' arrival time.

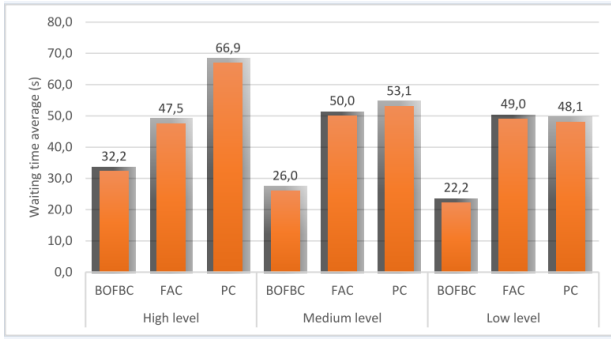
After choosing a two movements combination, the controller calculates the adaptive green signal duration using the adaptive equation eq.5. With h is the time of headway (supposed to be fixed), $N_{\langle ij \rangle}$ represents the amount of vehicles at the movement $\langle ij \rangle$, and t_f is the time taken by

the first vehicle to cross the intersection.

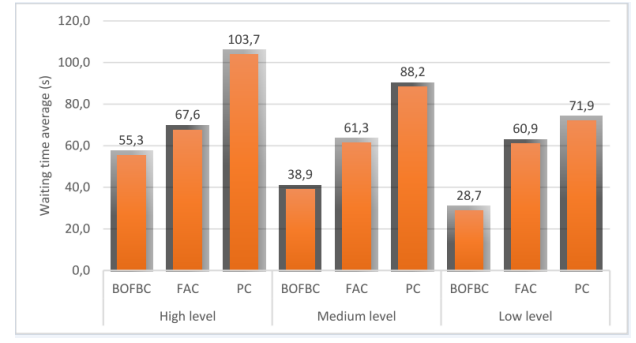
Using this approach, wasting time inter-phases (while there is no traffic over the cycle of the green signal) is minimized using an adaptive method for measuring the period of the green light. The number of vehicles considered to make a decision $N_{\langle ij \rangle}$ is exactly the number of vehicles allowed to cross the stop line.

$$t_{G\langle ij \rangle} = (N_{\langle ij \rangle} - 1) \times h + t_f \quad (5)$$

It is important to highlight that only a single movement could have the green light. This can be happened in the case of a low flow of traffic and all possible movements combinations are empty from vehicles. The green light cycle



(a)



(b)

Fig. 5: Average of waiting time in the network of four intersections: (a) half simulation period, (b) full simulation period

	Average emissions (mg/s)							
	At half simulation period				full simulation period			
	CO	CO2	HC	NOx	CO	CO2	HC	NOx
High level								
BOFBC	9762,6	379285,9	53,0	152,3	13714,9	446593,9	72,6	182,8
FAC	11975,1	411465,2	63,8	167,3	15425,2	472563,4	81,0	194,7
PC	15196,7	458021,0	79,6	189,2	21431,9	561684,4	110,4	236,4
Medium level								
BOFBC	8602,7	359376,9	47,3	143,3	10939,4	403635,2	58,9	162,9
FAC	12139,0	409597,5	64,5	166,8	14211,1	451965,1	75,0	185,4
PC	12903,9	421381,3	68,3	172,3	18874,4	524879,0	97,9	219,1
Low level								
BOFBC	8006,1	353548,2	44,4	140,2	9180,7	376096,6	50,3	150,2
FAC	11920,6	410063,2	63,6	166,7	14039,9	449654,8	74,1	184,3
PC	12083,4	412019,4	64,3	167,6	16181,2	483716,9	84,7	200,0

BOFBC: bi-objective function based control.

FAC: fully actuated control.

PC: pre-timed control.

TABLE I: Half and full simulation duration for four intersections: for the three traffic levels, SUMO emissions results for bi-objective-based-control Vs pre-timed and fully actuated control.

for the selected movements $\{x, y\}$ is determined as an output of this algorithm. Then, the external events $C_{\langle x \rangle}$ and $C_{\langle y \rangle}$ take 1 as a value ($x = \langle ij \rangle$ and $y = \langle i'j' \rangle$ with $i, i' \in \{N, S, E, W\}$, $j, j' \in \{s, l\}$, and $x \neq y$). As a result, these events would cause the firing of their associated transition $T_{6\langle x \rangle}$ and $T_{6\langle y \rangle}$ in the global TSPN model. Eventually, after their firing, the traffic lights become green and the red lights turns off for the next selected combination $\{x, y\}$.

IV. SIMULATION OUTCOMES AND DISCUSSIONS

As described above, our ultimate objective is to make traffic at urban intersections as smooth as possible by mitigating the queuing phenomenon and thereby reducing its negative impact on the drivers and the atmosphere. To this end, a comparative analysis is provided in this subsection between three control strategies. To this end, we compare the proposed bi-objective function based-control (BOFBC), the pre-timed control (PC), and the fully actuated control (FAC) (see

[14]). The well-known SUMO simulation tool [15] is used to represent the virtual model of the network and simulate its behavior as well as validate the proposed control logic. This virtual model is conceived thanks parameters of physical network and collected information from implemented sensors in the intersections. The suggested TSPN model and the control strategy are then taken into account in the virtual model.

a) Waiting time analysis

A high queuing of vehicles makes traffic networks more congested and consequently induces huge waiting times. We aim then to reduce these waiting times at each intersection of the network.

Fig.5 gives the average waiting time for each vehicle, for the three traffic levels, and for the half of simulation period. For the two measurement catches during the simulation period, it is clear that the BOFBC strategy represents the lower average waiting time compared with the FAC and

PC strategies. And this for the whole network of four intersections, for all traffic flow.

d) Gas emissions analysis

The considered environmental indicators in this paper are CO, CO₂, HC, and NO_x emissions. For the traffic levels of the first scenario, we perform a comparison of the three control approaches based on the same input data (route file). The results collected are summarized in Table I for two steps: at half and end simulation period. By analyzing the obtained results, we see that the proposed adaptive approach substantially minimizes gas emissions at the network and for all traffic levels.

V. CONCLUSION

This paper studied traffic flow at urban intersections with random behavior and adverse effects on the environment as a dynamic discrete event system. Our aims are to model the behavior of this process and establish an optimized adaptive control plan to ensure a smooth traffic, and to optimize both fuel consumption and volatile gas emissions. To do so, we proposed a methodology with two steps. First, we develop a timed synchronized Petri net based model to represent and studied the network behaviour. And second, we proposed a SUMO based simulation approach to model the network behaviour, check and validate its well-operating properties.

The theoretical modeling approach is based on a modular method enabling to represent each movement of the network independently. In addition, the concept of master-slave collaboration is considered to maintain the required functioning of the developed model and setup a dynamic control method for traffic regulation. The master (controller) uses a bi-objective function to control the traffic lights and make decisions for light switching according to the traffic status within the network. In accordance with the arrival times of vehicles and their number in each queue, the controller finds the combination of movements (without conflicts) for which the light will turn green for the next cycle. Also, it calculates the required time to evacuate all vehicles from these movements while ensuring a smooth and safe traffic, and avoiding vehicles delays.

The proposed model is checked and validated by simulation. To this end, we have used the well-known simulator SUMO as powerful tool for traffic simulation. Some of the proposed TSPN model's properties are checked and proven by simulation and the well-functioning of the studied network is confirmed by obtained results. In addition, the performed simulations demonstrate that the suggested control solution (BOFBC) substantially decreases the waiting time of vehicles within the intersections, fuel consumption, and gas emissions. These findings show that our built approach allows vehicles to cross the network (intersections) as quickly as possible compared to some existing control strategies in the literature (FAC and PC). Our primary goal now is to consider real traffic data collected using the IoT implemented into the intersections for the real time control of the cyber-physical system. Also, we intend to integrate unpredictable bottleneck of resources in the network. In other words,

to evacuate the network after eventual accidents and give priority to emergency vehicles using green waves for interconnected intersections. Finally, V2V (vehicle to vehicle) and V2I (vehicle to infrastructure) communications are also tracks to investigate in order to control and regulate road traffic flows within the intersections in urban areas.

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